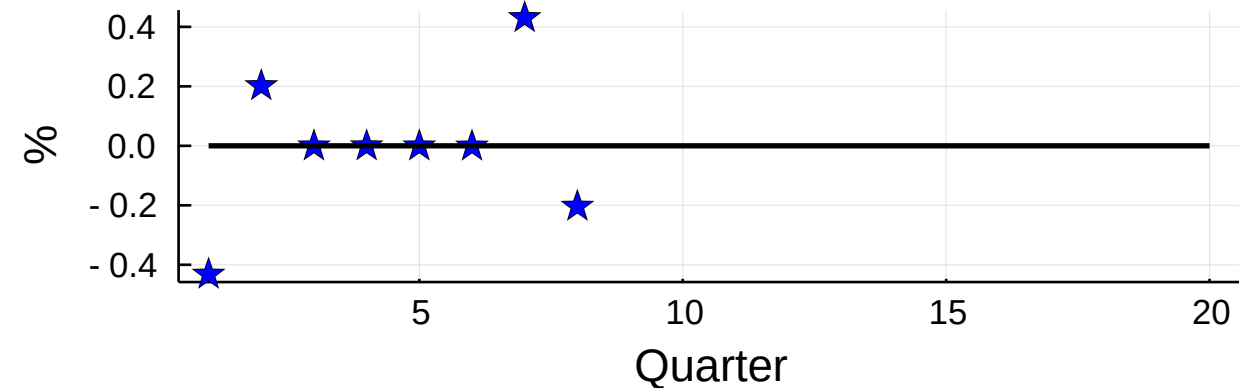
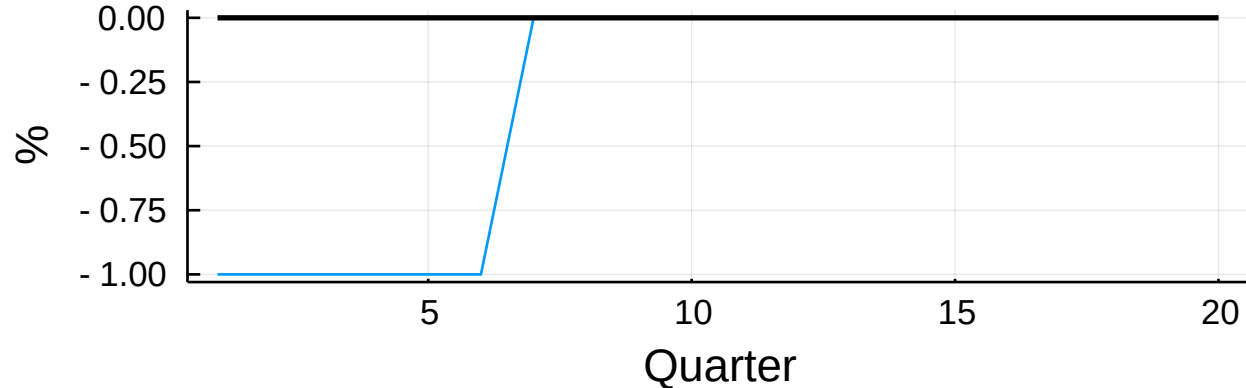


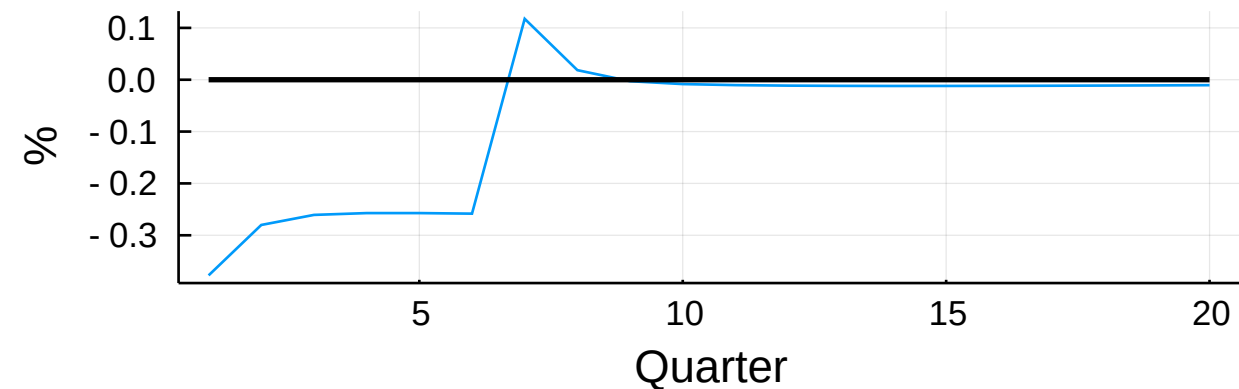
### Permanent Liquidity Innovations (APR)



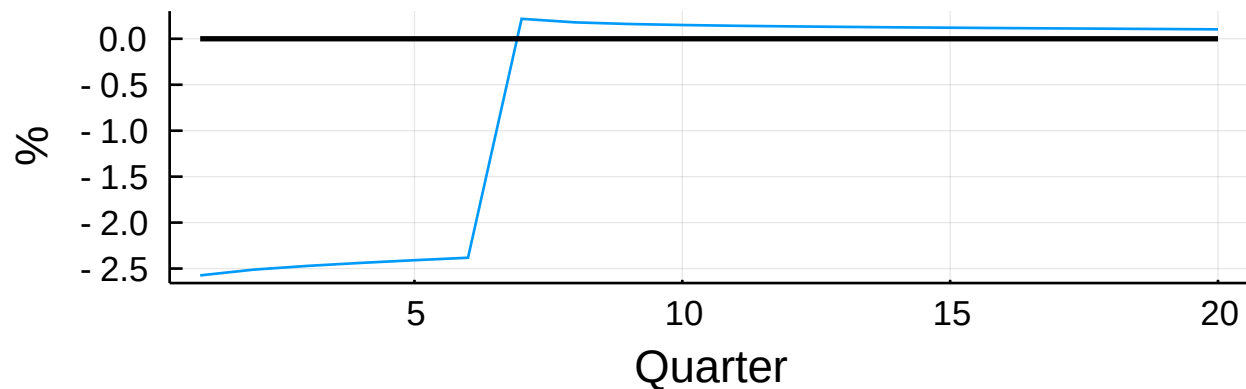
### Policy rate (APR)



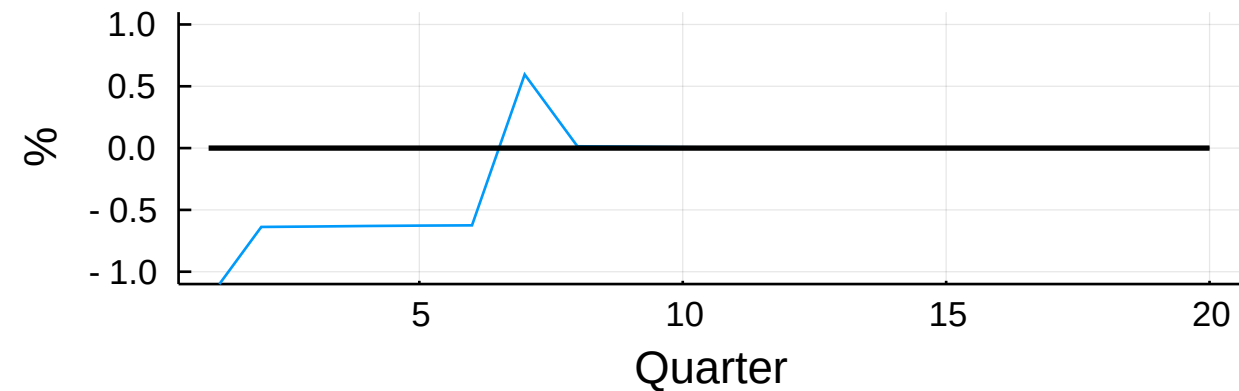
### Inflation (% , yoy)



### Output (% dev from SS)



### $r^*$ (Forward 5- year real natural rate, APR)



### Ex- ante real rate (APR)

